```
%%%Question 1%%%
  x = 1:100;
  y = find(mod(x, 2) == 0);
  fprintf("--------Question 1--------");
  fprintf("\n");
  fprintf("x:");
  fprintf("\n");
  disp(x);
  fprintf("\n");
  fprintf("y:");
  fprintf("\n");
  disp(y);
  fprintf("\n");
  clearvars;
%%%Question 2%%%
  t = [0, 0.1, 0.2, 0.3, 0.4];%Time in Seconds%
  f = [10, 15, 20]; %Frequency in Hz%
  A = zeros(5, 3);

  for i = 1:5
     for j = 1:3
        A(i,j) = 3 .* cos(2 .* pi .* t(i) .* f(j) + 0.1);
     end
  end
  fprintf("--------Question 2--------");
  fprintf("\n");
  fprintf("Frequencies: ");
  fprintf("%d ", f);
  fprintf("\n");
  fprintf("Times: ");
  fprintf("%.1f ", t);
  fprintf("\n");
  fprintf("result:");
  fprintf("\n");
  disp(A);
  clearvars;
%%%Question 3%%%
  iteration = 0.01;
  t = 0;
  A = [ 35, 40, 45; 0, 0, 0];
  Condition = 0;
  function[y] = f1(x,omega)
     y = exp(1.2) .*cos(omega.*x);
  end
  function[y] = f2(x)
     y = x.^3;
  end

  for i = 1:3
```

```
        while Condition < 10
            Condition = f2(t); % I would not know how to do this without knowing the functions
            if (10 > f1(t,A(1,i))) && (10 > f2(t))
                A(2,i) = t;
            end
            t = t + iteration;
        end
        Condition = 0;
        t = 0;
    end
    fprintf("--------Question 3--------");
    fprintf("\n");
    fprintf("Omega: ");
    fprintf("%d  ", A(1,:));
    fprintf("\n");
    fprintf("t max: ");
    fprintf("%.2f  ", A(2,:));
    fprintf("\n");
    clearvars;
%%%Question 4%%%
    t = linspace(0,1,15);

    x = 4 .* cos(2 .* pi .* t + 0.2) + 3 .* sin(pi.^2 .* t);

    x_avg = mean(x);

    x_max = max(x);

    x_min = min(x);

    fprintf("--------Question 4--------");
    fprintf("\n");
    fprintf(" Samples of 4cos(2pi *t + 2.0) + 3sin(pi^2 *t):");
    fprintf("\n");
    fprintf("%.6f   ", x);
    fprintf("\n");
    fprintf("Average of samples: %.6f",x_avg);
    fprintf("\n");
    fprintf("Max of samples: %.6f",x_max);
    fprintf("\n");
    fprintf("Min of samples: %.6f",x_min);
    fprintf("\n");

    clearvars;
%%%Question 5%%%
    t = 0:0.1:10;
    function[y] = s1(x)
        y = sin(2.*pi.*0.2.*x);
    end
```

```
function[y] = s2(x)
    y = sin(2.*pi.*0.425.*x + 0.4);
end
function[y] = s3(x)
    y = s1(x) + s2(x);
end
figure(1);
plot(t, s1(t), "Red");
hold on
plot(t, s2(t), "Blue");
hold on
plot(t, s3(t), "Green");
legend("s1",  "s2", "s3");
xlabel("t");
ylabel("S(t)");
title("Question 5");
hold off

clearvars;
%%%Question 6%%%

function[y] = MySinc(x)
    if x == 0
        y = 1;
    else
        y = sin(x) ./ x;
    end
end
x = linspace(-2.*pi, 2.*pi);
figure(2);
plot(x,sinc(x), "Black");% apparently the matlab sinc uses sin(pi x)/ pi x
hold on
plot(x, MySinc(x), "Magenta");
legend("sinc()","MySinc()");
title("Question 6");
```

```
>> Lab1_ECE103L
--------Question 1--------
x:
  Columns 1 through 31

     1     2     3     4     5     6     7     8     9    10    11    12    13    14    15    16    17    18    19    20    21    22    23    24    25    26    27    28    29    30    31

  Columns 32 through 62

    32    33    34    35    36    37    38    39    40    41    42    43    44    45    46    47    48    49    50    51    52    53    54    55    56    57    58    59    60    61    62

  Columns 63 through 93

    63    64    65    66    67    68    69    70    71    72    73    74    75    76    77    78    79    80    81    82    83    84    85    86    87    88    89    90    91    92    93

  Columns 94 through 100

    94    95    96    97    98    99   100


y:
  Columns 1 through 31

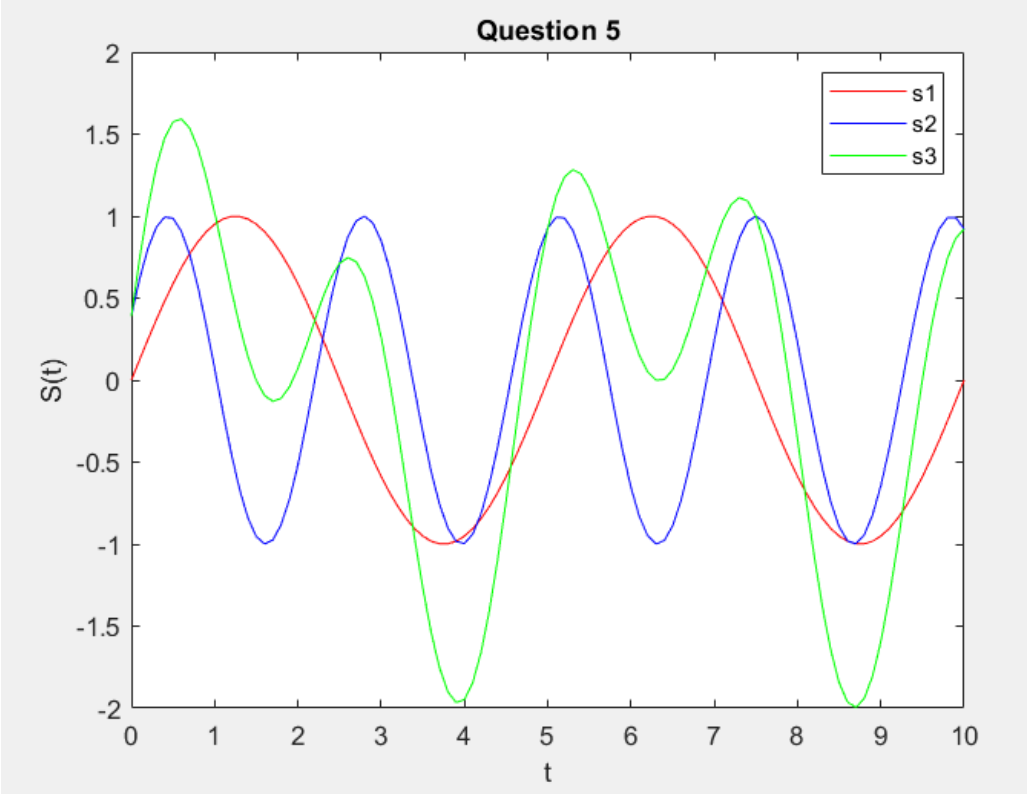
     2     4     6     8    10    12    14    16    18    20    22    24    26    28    30    32    34    36    38    40    42    44    46    48    50    52    54    56    58    60    62

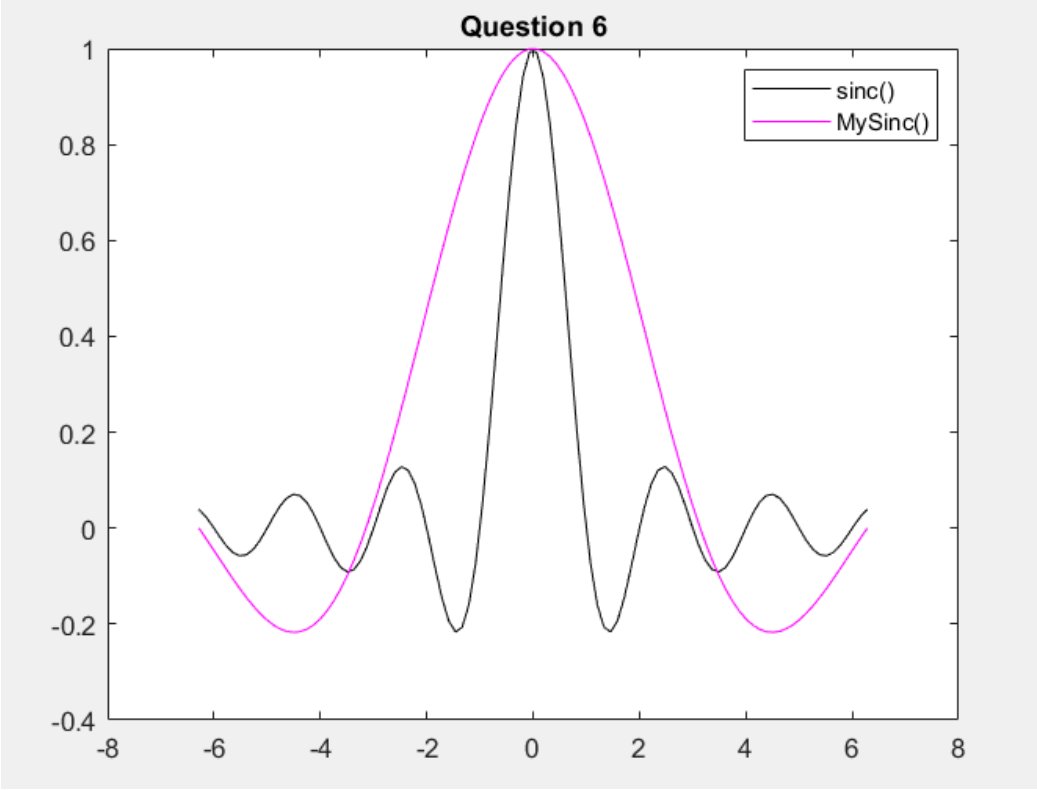
  Columns 32 through 50

    64    66    68    70    72    74    76    78    80    82    84    86    88    90    92    94    96    98   100


--------Question 2--------
Frequencies: 10 15 20
Times: 0.0 0.1 0.2 0.3 0.4
result:
    2.9850    2.9850    2.9850
    2.9850   -2.9850    2.9850
    2.9850    2.9850    2.9850
    2.9850   -2.9850    2.9850
    2.9850    2.9850    2.9850

--------Question 3--------
Omega: 35  40  45
t max: 2.15  2.15  2.15
--------Question 4--------
 Samples of 4cos(2pi *t + 2.0) + 3sin(pi^2 *t):
3.920266   5.131277   4.784215   2.664339   -0.698539   -4.187404   -6.534267   -6.846370   -4.987025   -1.638378   1.983336   4.632318   5.531896   4.648498   2.629363
Average of samples: 0.735568
Max of samples: 5.531896
Min of samples: -6.846370
```


Question 5
s1
s2
s3
S(t)
t
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
1
2
3
4
5
6
7
8
9
10


Question 6
sinc()
MySinc()
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-8
-6
-4
-2
0
2
4
6
8